

REFERENCE CARD

Git Credential Manager Fresh Linux Machine Setup

Everything Git Bash gives you for free on Windows — GCM installed, configured, and wired to a persistent credential store — reproduced step-by-step on Ubuntu/Debian.

[Ubuntu / Debian](#)[Desktop session](#)[GitHub + GitLab](#)

1 Download the latest GCM .deb for your architecture

Pulls the current release tag from GitHub automatically — no version number to hardcode or go stale.

```
GCM_VERSION=$(curl -s https://api.github.com/repos/git-ecosystem/git-credential-manager/releases/latest \
  | grep '"tag_name"' | cut -d'"' -f4 | sed 's/^v//')

curl -sL -o gcm-linux-x64-${GCM_VERSION}.deb \
  "https://github.com/git-ecosystem/git-credential-manager/releases/download/v${GCM_VERSION}/gcm-linux-x64-${GCM_VERSION}.deb"
```

Note: swap `x64` for `arm64` on ARM machines (e.g. Raspberry Pi, Apple Silicon VMs).

2 Install the package

```
sudo dpkg -i gcm-linux-x64-${GCM_VERSION}.deb
# only if dpkg complains about missing dependencies:
sudo apt-get install -f
```

3 Register GCM as git's credential helper

```
git-credential-manager configure
```

4 Use browser-based login, not device code

```
git config --global credential.https://github.com.GitHubAuthModes browser
```

5 Pick a persistent credential store

Check whether a desktop keyring's Secret Service is already running:

```
busctl --user list 2>/dev/null | grep -i secret
# look for: org.freedesktop.secrets (GNOME Keyring / KWallet)
```

✓ Found it — typical on any GNOME / KDE desktop session

```
git config --global credential.credentialStore secretservice
```

✗ Not found — headless server / WSL / container

```
sudo apt-get install -y gpg pass
gpg --quick-generate-key "$(git config user.email)" default default never
git config --global credential.credentialStore gpg
```

6 Trigger the one-time browser authentication

```
git push # or `git fetch` in any repo — this pops the browser login
```

Log in once in the browser → GCM caches the token in whichever store you picked → every push/pull after that is **silent, permanent** — identical to how Git Bash behaves on Windows.

Verify it's wired up correctly

```
git config --global --get-all credential.helper
git config --global --get credential.credentialStore
git config --global --get credential.https://github.com.GitHubAuthModes
```

Why Windows needs none of this

Aspect	Windows (Git Bash)	Linux (this guide)
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GCM install	Bundled inside the Git for Windows installer since 2.28 (2020)	Separate <code>.deb</code> download + <code>dpkg -i</code>
Credential store	Only one sane option → Windows Credential Manager, wired automatically	Multiple valid options (Secret Service, gpg/pass, plaintext) → must choose explicitly
Steps you see	Just the one-time browser login	Steps 1–5 once per machine, then identical one-time browser login

One-time setup per machine — not a per-repo or per-push chore.